

ZELVIANA PUTRI

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PROFILE

Master's candidate in **Advanced Bioconvergence** with interdisciplinary research experience integrating **metabolomics, bioinformatics, analytical chemistry, and cellular biology** to identify bioactive compounds and investigate their molecular mechanisms relevant to human health. Skilled in **chromatographic and spectroscopic techniques** (Prep-HPLC, LC-MS, MPLC, TLC, and 1D/2D NMR analysis), **computational approaches** (molecular docking, and network pharmacology), and **biological assays** including antimicrobial, cell viability, and other in vitro/in vivo assays. Aspires to become a **translational biomedical researcher** by integrating omics technologies, computational biology, and biotechnology to advance therapeutic discovery.

EDUCATION

M.Sc. Candidate in Advanced Bioconvergence | CGPA: 4.30 / 4.30 **09/2024-08/2026**

Kyungpook National University, South Korea

- **Relevant Courses:** Advanced Food Chemistry, Advanced Natural Product Resources, Natural Product Biofunctionality, Metabolomics in Biotechnology, Advanced Physiology, Drug and Food Nutritional Genomics
- **Thesis:** Discovery of Longevity-Promoting Compound from Hop (*Humulus lupulus*) Integrating Targeted Metabolomics, Network Pharmacology, Molecular Docking, and *Caenorhabditis elegans* Validation
- Fully funded **Global Korea Scholarship (GKS)** recipient awarded by the National Institute for International Education (NIIED), Ministry of Education, Republic of Korea

Korean Language Study

09/2023-08/2024

Kyungpook National University, South Korea

One-year intensive Korean language program funded by the Global Korea Scholarship (GKS). Awarded **Best Student (Level 3)** and achieved **TOPIK Level 3** (94th Test)

B.Sc. in Food Technology | CGPA: 3.43 / 4.00

08/2015-06/2019

Universitas Brawijaya, Indonesia

- **Thesis:** Microemulsification of Crude Palm Oil Containing Multi-Components Bioactive Compounds: The Types of Emulsifiers and Concentration of Oil Fraction
- **Laboratory Teaching Assistant:** Basic Physics, Microbiology, Biochemistry and Food Analysis, Nutrition and Food Evaluation (2016–2018)
- **Research Grant & Scholarship Recipient**, Ministry of Research, Technology and Higher Education, Indonesia

RECENT RESEARCH EXPERIENCES

Graduate Researcher

09/2024-06/2026

Department of Advanced Bioconvergence, Kyungpook National University

- Investigated secondary metabolites from *Humulus lupulus*, *Ananas comosus* peel, and *Aquilaria malaccensis* for their bioactive properties and potential relevance to health.
- Developed extraction, chromatographic separation, and purification workflows using MPLC and prep-HPLC, with LC-MS and 1D/2D NMR for structural characterization.
- Successfully isolated four compounds of *Humulus lupulus*, six derivative compounds of *Ananas comosus* peel, and two compounds of *Aquilaria malaccensis*
- Integrated molecular docking and network pharmacology to investigate the potential molecular mechanisms underlying the biological activities of those purified compounds

Graduate Researcher (Collaborative Research)

04/2026-06/2026

Physio-Genomics Laboratory, Kyungpook National University

- Established non-cytotoxic concentration ranges of lupulone C using cell viability assays

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- Evaluated anti-inflammatory activity by quantifying nitric oxide (NO) production in LPS-stimulated RAW 264.7 macrophages

Graduate Researcher (Collaborative Research)

11/2025-01/2026

Food Metabolic Engineering Laboratory, Kyungpook National University

- Designed and conducted lifespan assays using *Caenorhabditis elegans* to evaluate the longevity-promoting potential of *Humulus lupulus*'s secondary metabolites
- Evaluated the antibacterial activity of *Ananas comosus* peel extracts

RESEARCH OUTPUTS

Putri, Zelviana., Soo Rin, Kim, Dongyup, Hahn (2026). Discovery of Longevity-Promoting Compound from Hop (*Humulus lupulus*) Integrating Targeted Metabolomics, Network Pharmacology, Molecular Docking, and *Caenorhabditis elegans* Validation. Manuscript in preparation.

Putri, Z., & Hahn, D. (2025). Isolation, structural elucidation, and molecular mechanisms underlying the anti-aging effects of xanthohumol from hops using targeted metabolomics, network pharmacology, and in vitro studies. KoSFoST International Symposium and Annual Meeting, Korea, July 2–4, 2025.

Jauhar, M. M., Wafqiah, A., Syaifie, P. H., Mardiyanti, E., **Putri, Zelviana.**, & Edi, R. Y. (2023). Komposisi Herbal untuk Suplemen Vitalitas Pria [Herbal Composition for a Male Vitality Supplement]. Indonesian Patent Application No. P00202314238.

Putri, Zelviana., Syaifie, P. H., Siregar, K. A. K., Ibadillah, D., Cintya, D. W., Jauhar, M. M., Arda, A. G., & Mardiyati, E. (2023). Exploring the synergistic and anti-diabetic effects of new nutraceutical combination of *Phyllanthus niruri*, *Momordica charantia*, *Cinnamomum verum* and propolis by network pharmacology, molecular docking, and experimental studies. 4th International Conference on Applied Molecular and Pharmaceutical Sciences (ICAMPS), Nusa Dua, Bali, Indonesia, August 3, 2023.

Fadhilah, Nidya., **Putri, Zelviana.**, Qodariyah, N. L., Rahmawani, D. E., & Atrinto, A. C. P. (2017). B-SMART: (Bag of Silica with Mahogany Extract) using chemical method of ultrasound as an effort to improve the quality of rice. Proceedings of the National Agricultural Technology Symposium, May 17, 2017, Brawijaya University, Malang, Indonesia, pp. 143–150. ISBN 978-602-74352-1-6.

OTHER RECENT RESEARCH CONTRIBUTIONS

Isolation and characterization of Pineapple peel's Secondary metabolites **2025**

Kyungpook National University, South Korea

- Upcycled *Ananas comosus* peel by-products by recovering and characterizing bioactive secondary metabolites.
- Selected for the BK21 Graduate Research Competition

Isolation, Purification and characterization of agarwood (*Aquilaria malaccensis*) **2025**

Kyungpook National University, South Korea

Discovered and structurally characterized secondary metabolites from *Aquilaria malaccensis* using **MPLC, semi-preparative HPLC, LC-MS, and 1D/2D NMR**.

TEACHING EXPERIENCES

Laboratory Assistant for Nutrition and Food Evaluation

09-12/2018

Brawijaya University, Malang – Indonesia

- Conducted experiments and developed research method collaborated with the lecturers
- Successfully assisted and taught 200+ junior students to do experiments
- The material courses were Starch Nutrition Value, Food Fiber, and Protein Nutrition Value Evaluation, Determining Tannin and Phytate Content, and Experiment Animals

Laboratory Assistant for Biochemistry and Food Analysis**02-06/2018***Brawijaya University, Malang – Indonesia*

- Prepared chemical solutions, reagents, and raw materials, the laboratory tools
- Successfully assisted and taught 200+ junior students to do experiments
- The material courses were Amylase Enzyme Extraction and Activity Test, Water and Ash Content, Fat Protein, and Vitamin C Analysis

Brawijaya University, Malang – Indonesia**09-12/2016***Laboratory Assistant for Microbiology*

- Conducted experiments and developed research method collaborated with the lecturers
- Prepared chemical solutions, reagents, and raw materials, the laboratory tools
- Successfully assisted and taught 200+ junior students to do experiment
- The material courses were Aseptic and Sterilization Methods, Isolation, Cultivation and Culture Preservation, Microbe Enumeration Techniques

RELEVANT WORK EXPERIENCES**PT Nanotech Indonesia Global, South Tangerang - Indonesia**

PT Nanotech Indonesia Global is a Tech Company whose core business is in the form of science and technology services based on research and development (R&D), material engineering, and nanotechnology

Link: <https://nig.co.id/>

*Junior Researcher at Nano Center Indonesia***12/2022-8/2023**

A private nanotechnology research start-up under PT Nanotech Indonesia Global that focus on synthesising nanomaterials and their applications. Consist of Applied Physics and Chemistry, Advanced Materials, Applied Technology, and Life Science. **Link:**

<https://www.nanocenter.id/>

- Conducted researches related to food technology and nutrition, functional food, and nutraceutical
- Worked on multiple compounds for therapeutic targets using a bioinformatics approach (network pharmacology and in silico study) and in vitro validation
- Successfully discovered a novel nutraceutical for Type 2 diabetes mellitus (T2DM)
- Successfully produced **a research patent** of herbal supplement composition for men's vitality named "Komposisi Herbal untuk Suplemen Vitalitas Pria"
- Published **a book** "Kajian Potensi Produk Hasil Budidaya Lebah Trigona Sumatera Barat: Propolis dan Madu" collaborated with **The Ministry of Environmental and Forestry, West Sumatera Province**

*Research and Development Staff at Nanotech Natura Indonesia***12/2022-8/2023**

A spin-off from Nano Center Indonesia which focuses on cosmetic and health products based on natural resources and nanotechnology. **Link:** <https://nanotechnatura.com/>

- Discussed with clients about their requested product qualifications
- Analysed and developed new product formula to meet client's inquiry
- Conducted trial and error of the new product formula

Ajinomoto Indonesia., Ltd. Mojokerto –Indonesia**01 – 02 / 2018**

One of the largest International Fast-moving Consumer Goods in Indonesia that produces high quality products like Monosodium Glutamat (MSG), Seasoning, etc.

<https://www.ajinomoto.com/>

Internship Program

- Learned the production process of MSG and implemented the GMP and HACCP's procedure
- Made a report to the supervisor regarding the actual process in the factory
- Successfully achieved A+ grade score for the internship program

NOTABLE ACHIEVEMENTS

Best Research Poster at Korean Society of Food Science and Technology (KoSFoST) in 2025 - Korea	07/2025
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An annual International Conference by Korean Society of Food Science and Technology. "Isolation, structural elucidation, and molecular mechanisms underlying the anti-aging effects of xanthohumol using targeted metabolomics, network pharmacology, and in vitro"

Research Grant Recipient at BK21 Research Graduate Competition in 2025 - Korea	2025-2026
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An annual Research Graduate Competition by Brain Korea 21 (BK21). Proposed "Integrated Upcycling of Pineapple Peel Waste through Isolation of Bioactive Natural Compounds and Biochar Production"

Gold Medal at Korea International Women's Invention and Exposition - Korea	06/2019
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An annual International research competition by Korea International Inventor Association. Proposed "Analog Rice from Tapioca and Sorghum Flour with Fortification of Moringa Leaves Flour to Combat Malnutrition in Realizing SDGs"

1st Winner at National Food Technology Competition - Indonesia	05/2019
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An annual National research competition by Katolik Widya Mandala University. Proposed "Innovation of Synbiotic Yogurt Jelly Drink Made from Local- based Food to Combat Colorectal Cancer (CRC) in Indonesia"

Participant at Institute of Food Technologists Student Association Competition - USA	10/2018
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An annual International research competition by Institute of Food Technologists. Proposed "Instant Sweet Porridge to Combat Vitamin A Deficiency of Kenyan Preschool Children (baby corn, baby carrot, and banana peels)"

INTERNATIONAL CONFERENCES

• Korea Metabolomics Society 14 th Annual Meeting – South Korea	04/2026
• [Poster Presentation] 52 nd Annual Meeting & International Symposium – South Korea	01/2026
• [Poster Presentation] 13 th KSP-JSP-CPA Joint Symposium – South Korea	11/2025
• [Best Poster] Korean Society of Food Science and Technology (KoSFoST) – South Korea	07/2025
• [Poster Presentation] Korea Metabolomics Society 13 th Annual Meeting – South Korea	04/2025
• [Poster Presentation] 14 th International Congress of Diabetes and Metabolism – South Korea	10/2024
• [Oral Presentation] 10 th International Conference of Sensors in conjunction with the 4 th International Conference on Advanced Materials and Practical Nanotechnology – Indonesia	08/2023
• Global Agri-Food Scientific Symposium - Singapore	10/2022
• [Product Exhibition] Korea International Women's Invention and Exposition - South Korea	06/2019
• [Product Exhibition] International Invention and Innovative Competition - Malaysia	10/2018
• [Best Paper] 1st Young Scientist International Seminar, Brawijaya University - Indonesia	07/2017
• [Product Exhibition] National Agricultural Technology Symposium - Indonesia	07/2017

PERSONAL & TECHNICAL SKILLS

• PyMol	• PyRex	• Basic Python	• ChemDraw
• Autodock Vina 1.1.2	• Autodock Tools 1.5.7	• Cytoscape	• Minitab
• Airtable	• MestReNova	• Google workspace	• SPSS
• Research and Development	• Analytical Thinking	• Public Speaking	• Leadership

LANGUAGES

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- Bahasa Indonesia (Mother Tongue)
 - English (IELTS 6.5) (Expired 02/2025)
 - Korean (Level 3)
 - Germany (Currently studying)

REFERENCES

1. Prof. Dongyup Hahn, Ph.D.

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2. Prof. Soo Rin Kim, Ph.D

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3. Assisstant Prof. Joe Son, Ph.D

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